PG No. 17 $C_{3\mathrm{i}}$ $\bar{3}$ [trigonal]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$
E_g	$1:\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$-1:\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$
E_u	$1:\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$-1:\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & \frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{bmatrix}$